

Case Study 1: Cyclistic Bike-share

Tre'Shunda Jmaes

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1 Introduction

In 2016, Cyclistic launched a successful bike-share offering. Since then, the program has grown to a fleet of 5,824 bicycles that are geotracked and locked into a network of 692 stations across Chicago. The bikes can be unlocked from one station and returned to any other station in the system anytime. Until now, Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the flexibility of its pricing plans: single-ride passes, full-day passes, and annual memberships. Customers who purchase single-ride or full-day passes are referred to as casual riders. Customers who purchase annual memberships are Cyclistic members. Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, Moreno believes that maximizing the number of annual members will be key to future growth. Rather than creating a marketing campaign that targets all-new customers, Moreno believes there is a solid opportunity to convert casual riders into members. She notes that casual riders are already aware of the Cyclistic program and have chosen Cyclistic for their mobility needs. Moreno has set a clear goal: Design marketing strategies aimed at converting casual riders into annual members. In order to do that, however, the team needs to better understand how annual members and casual riders differ, why casual riders would buy a membership, and how digital media could affect their marketing tactics. Moreno and her team are interested in analyzing the Cyclistic historical bike trip data to identify trends.

2 Business Problem (Ask)

In this study, I explore how do annual members and casual riders use Cyclistic bikes differently?

3 Data Sources (Prepare)

The data used in this study is publicly available at <https://divvy-tripdata.s3.amazonaws.com/index.html>. The sample data set encompass the most re-

cent 12 months of trip data (June 2024 - May 2025). The data include: ride id, rideable type, start time, start station name, end time, end station name, start station id, end station id, start longitude, start latitude, end longitude, end latitude, and rider type. After downloading and organizing the data, the files were opened and explored in Python. The data has been made available by Motivate International Inc. under this license. Data-privacy issues prohibits using riders' personally identifiable information. This means that I was unable to connect pass purchases to credit card numbers to determine if casual riders live in the Cyclicistic service area or if they have purchased multiple single passes.

4 Data Cleaning (Process)

To process the data, I used SQL. In SQL I merged each individual file into one large file (I could have also done this in Python). To ensure the data was clean, I checked for duplicates, missing data, and consistent data formatting before merging the individual files. I noted that there is some missing station information, otherwise the data was fairly clean. The missing station information was interpreted as “Null” values and filtered out in the analysis when relevant.

5 Analysis (Analyze)

In SQL, I investigated the difference between the duration of rides between causal riders and members. To do so, I grouped the data by member type and rideable type, then calculated the number of riders in each group and the total duration of the ride for each group. To answer, which stations do each rider type use most, I queried the data, filtering out “Null” values for the station name, for the top 10 most frequently used start and end stations. I investigated what was the busiest weekday among riders by grouping the data by weekday and calculating the total number of riders. Additionally, I filtered the data by season to see how the results changed over season.

6 Results and Insights (Share)

I find that, over the past 12 months:

1. Members and casual riders have ridden electric bikes most often.
2. Members ride more often than casual riders.
3. Casual riders ride classic bikes for longer than members.
4. Casual riders use the station at Streeter Dr & Grand Ave. the most.
5. Members use the station at Kingsburg St & Kinzie St. the most.
6. Casual riders ride mostly on Saturdays which are the busiest days.

7. Members mostly ride on Thursdays.

For every season, individually:

1. The majority of the riders are members.
2. Electric bikes are ridden the most overall and by both groups.
3. The summer is the busiest season, with both members and casual riders riding the most during that time.
4. Casual riders end at Streeter Dr & Grand Ave most.
5. Members end at Kingsbury & Kinzie St most.
6. Casual riders ride mostly on Saturdays.

During the Summer (June, July, August):

- Classical bikes are ridden the longest by casual riders and members.
- Casual Riders start Streeter Dr & Grand Ave most.
- Members start at Broadway & Barry Ave most.
- Members ride mostly on Wednesdays.
- The busiest days are Saturdays.

During the Fall (September, October, November):

- Members ride electric bikes the longest.
- Casual Riders start Streeter Dr & Grand Ave most.
- Members start at Canal St & Madison St the most.
- Members ride mostly on Wednesdays.
- The busiest days are Wednesdays.

During the Winter (December, January, February):

- Members ride electric bikes the longest.
- Members and casual riders start at Kingsbury & Kinzie St most.
- Casual Riders start at Streeter Dr & Grand Ave most.
- Members start Kingsbury & Kinzie St most.
- Members ride mostly on Tuesdays.
- The busiest days are Tuesdays.

During the Spring (March, April, May):

- Members ride electric bikes the longest.
- Members start Kingsbury & Kinzie St most.
- Casual Riders start Streeter Dr & Grand Ave most.
- Members ride mostly on Thursdays.
- The busiest days are Fridays.

The interactive visualizations are available here, on Tableau Public.

7 Recommendations (Act)

From the results provided in this study and the stakeholders goal of converting more casual riders into members, I recommend the stakeholders implement a marketing strategy in which target the most used stations by casual riders and the most used rideable by casual riders during the season in which casual riders ride the most. My top three recommendations are:

1. Implement advertisement for memberships at and around Streeter Dr & Grand Avenue.
2. Increase advertisement or promotions for memberships just before Summer.
3. Place advertisement on electric bikes.

Pricing information about memberships and different passes offered could be helpful to determine how spending trends between the groups differ and further inform the marketing strategy.

8 Conclusion

In conclusion, casual riders differ from members in some important ways. In the previous 12 months, casual riders most frequently used Streeter Dr & Grand Avenue station, road mostly during the Summer, and preferred the use of electric bikes. The recommendations presented here highlights the differences between members and casual riders that can help inform the marketing strategy in converting casual riders to members.

A Code

All code is available on GitHub at https://github.com/txj4522/case_study/tree/cyclist_bike_share